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To cite this article: Yiwei Hua, Ru Wang, Zuoxu Wang, Guoxin Wang & Yan Yan (09 Apr 2024): Knowledge graph with deep reinforcement learning for intelligent generation of machining process design, Journal of Engineering Design, DOI: [10.1080/09544828.2024.2338342](https://doi.org/10.1080/09544828.2024.2338342)

To link to this article: <https://doi.org/10.1080/09544828.2024.2338342>



Published online: 09 Apr 2024.



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Knowledge graph with deep reinforcement learning for intelligent generation of machining process design

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ABSTRACT

In the manufacturing industry, the design of machining processes plays a pivotal role in determining the quality, efficiency, and cost of product manufacturing. Machining process design is progressing towards intelligence to meet the high demands for efficiency and effectiveness in smart manufacturing. Building upon traditional computer-aided process planning, intelligent technologies, such as machine learning and knowledge graphs, have emerged as key drivers in advancing intelligent process design. To address knowledge accumulation, inflexible reuse, and fragmented reasoning in machining process design, this paper organically integrates the structured representation characteristics of the knowledge graph and the perceptual reasoning capabilities of deep reinforcement learning. It introduces an intelligent generation method for machining process design based on knowledge graph and deep reinforcement learning, aiming to achieve unified representation and reasoning reuse for historical process cases and general process rules. The effectiveness of the proposed method is validated through a case study involving the generation of machining process solutions for a specific model of diesel engine components. This research contributes valuable insights to overcoming the limitations of traditional methods and enhancing the efficiency of the machining process design.

ARTICLE HISTORY

Received 28 December 2023
Accepted 30 March 2024

KEYWORDS

Machining process design; knowledge graph; deep reinforcement learning; knowledge representation and reuse

1. Introduction

Efficient and high-quality manufacturing process design solutions are crucial. Specifically, machining process design plays a pivotal role in the link between product design and machining production and is a significant means for forming product components. As a core element in the manufacturing process, machining process design determines the sequence of machining operations (such as milling, drilling, and turning), processing methods, process parameters, equipment, and fixtures for a given part. Hence, the design of the

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machining process directly impacts manufacturing quality, efficiency, and cost (Yusof and Latif 2014). According to statistical data from an aircraft manufacturing company, the time allocated to process analysis and machining process planning constitutes over 70% of the entire machining activities for aircraft structural components, whereas the actual machining time is less than 30% (Huang et al. 2019a). Given the increasingly demanding nature of complex product machining tasks and the continuous improvement of machining quality, there is a pressing need for the development of intelligent process design to meet the high requirements of smart manufacturing for efficiency and effectiveness.

Since Niebel introduced Computer-Aided Process Planning (CAPP) in 1965, computers have replaced mainly manual process planning methods. Compared to manual design, CAPP significantly enhances the capability to handle complex process planning. Serving as a bridge between Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM), CAPP transforms product designs into optimal manufacturing sequences and allocates manufacturing resources rationally (Xu, Wang, and Newman 2011). With the advancement of technology, Al-Wswasi, Ivanov, and Makatsoris (2018) conducted a review on intelligent automated CAPP techniques. Adapa and Jagadish (2023) provided a comprehensive review of CAPP supported by feature-based, knowledge-based, and artificial neural networks. They discussed and analyzed generative and variant CAPP. Furthermore, numerous techniques and methods have emerged to enhance the effectiveness and intelligence of CAPP systems. For instance, Dou et al. (2023) proposed a genetic algorithm with path relinking for process planning in CAPP. Additionally, machine-learning-based approaches have been progressively proposed to enhance the intelligence of process planning systems (Kumar 2017). Yan and Melkote (2023) employed a deep generative model and siamese neural networks approach for automated manufacturability analysis and machining process selection, offering valuable methodological insights for machining process plan generation.

Researchers in the field have been dedicated to exploring and developing intelligent process design technologies and methods to achieve intelligent process design (Kumar 2019). Typical process design methods include optimisation-based approaches, process reuse methods, and rule-based reasoning methods (Wen et al. 2022). Optimisation-based systems typically treat the planning of machining process routes as an optimisation problem, aiming to minimise processing costs and enhance processing efficiency. However, these methods face challenges in knowledge accumulation and learning, hindering the extraction of generalised insights from optimizations of specific problems (Al-Wswasi, Ivanov, and Makatsoris 2018). In process reuse methods, historical process cases are often selected and reused to assist in machining process design. Nevertheless, these methods are limited in flexibility due to the relatively rigid nature of the process content itself (Wen et al. 2022). Rule-based reasoning methods infer the machining process through predefined process rules. However, these methods primarily focus on local and fragmented process reasoning, making it challenging to address macro-level process design issues (Zhang et al. 2022).

In recent decades, the rapidly advancing field of artificial intelligence has become a prominent research topic for addressing industrial manufacturing challenges. Particularly, Knowledge Graph (KG), as a vital research direction in the development of artificial intelligence towards cognitive intelligence, has demonstrated its intelligence advantages over traditional knowledge-based expert systems. This is evident in their applications in

intelligent reasoning, decision-making, semantic search, intelligent question-answering, recommendation systems, semantic understanding, and other scenarios (Zhang et al. 2021). Simultaneously, as a critical component of smart manufacturing, machining process design encompasses many historical cases and scattered general rules. It involves extracting process knowledge from process document data to fulfil the requirements of machining process planning for similar parts, operations, or features. It, in turn, reduces process planning time, effectively improving production efficiency and processing quality (Kumar 2019). Machining process design exhibits considerable reuse of experiential knowledge, particularly in knowledge-intensive tasks such as designing machining process solutions. Therefore, maximising the role of process knowledge in intelligent process design is crucial, necessitating accurate representation and efficient utilisation of machining process knowledge (Li et al. 2018).

Li et al. (2018) provided a comprehensive review of the application of knowledge representation methods in mechanical machining process planning. Wen, Ma, and Wang (2023) introduced a methodology focused on process knowledge for constructing a knowledge graph system to develop a domain ontology for process knowledge and facilitate process planning through knowledge-based question-answering mechanisms. However, the conventional management service framework for knowledge graph retrieval, recommendation, and question-answering fails to address the requirements of intricate and adaptable process plan formulation. Xiao et al. (2023) summarised the applications and advantages of knowledge graph in process design. These advantages include the ability to conveniently store and process complex process knowledge in multiple modalities, reveal implicit relationships among process knowledge, interconnect isolated process knowledge, and achieve globally optimal results by combining graphs to generate all process knowledge.

To bolster the capacity for inferring design intent across varying levels of process knowledge granularity, Huang et al. (2023) achieved the extraction of intent from operational steps, represented by tools, using deep learning and syntactic parsing. This contribution offers insights into reusing intricate process knowledge for inference purposes. However, deep learning is hampered by challenges such as the high cost of data annotation and limited interpretability. As a result, deep reinforcement learning has garnered increasing attention from scholars. While numerous studies have explored the intersection of deep reinforcement learning and process planning (Li et al. 2023; Zhang et al. 2024), research integrating knowledge graphs with deep reinforcement learning remains nascent.

In summary, extensive research has been conducted on intelligent design techniques and methodologies for machining technology. Progress has been achieved in matching reuse based on historical cases and local reasoning derived from fragmented rules. However, challenges persist in effectively leveraging case and rule knowledge for the intelligent generation of process solutions. Specifically, issues arise from inflexible reuse of case knowledge and fragmented reasoning of rule knowledge. These challenges hinder the unified organisation and efficient reuse of machining process knowledge in response to the demand for intelligent solutions. This paper proposes an intelligent generation method for machining process design solutions based on knowledge graphs and deep reinforcement learning. This method leverages the knowledge graph's structured representation capability and deep reinforcement learning's perceptual reasoning capability. It aims to achieve unified representation and reasoning reuse of a large amount of historical case knowledge and general rule knowledge in machining process design. Firstly, the characteristics

of comprehensive case knowledge and flexible rule knowledge are seamlessly integrated and mapped into the knowledge representation space model based on the ternary structure. Unlike traditional ontology, rule, or logic representations, this spatial representation can more effectively convey the topological information of process knowledge, which is vital for subsequent deep reinforcement learning model construction and process solution generation. Secondly, the framework for generating solutions for deep reinforcement learning based on knowledge graphs is established. Leveraging the advantages of deep reinforcement learning without requiring labelled training data, the inherent logic of the spatial knowledge representation model is incorporated into the model's reward settings, thereby guiding and expediting the model's training and learning processes. Finally, intelligent generation of machining process design solutions tailored to specific requirements is undertaken to facilitate the storage and sharing of knowledge subgraphs. Illustrated through a case study involving the generation of machining design solutions for diesel engine parts, the effectiveness of the proposed method is demonstrated, achieving intelligent reuse of existing process design case knowledge and rule knowledge through solution generation.

The paper makes three principal contributions. Firstly, it proposes a method for representing process design cases and rules in a unified manner. This method effectively captures the inherent connections among process design requirements, cases, rules, and scenarios, influencing the generation of process scenarios in a computable format. Secondly, it establishes an exploratory deep reinforcement learning framework grounded in knowledge graphs within process design. This framework defines actions as individual steps in generating relevant knowledge nodes and relationships within the process programme, with states representing the structural and content-related aspects of the overall process programme. Additionally, it designs a reward mechanism primarily based on case and rule knowledge, enabling the framework to generate a process programme that aligns with process rules and closely resembles existing machining process programmes. Finally, this method addresses the challenges associated with inflexible case knowledge reuse and fragmented rule knowledge reasoning observed in existing approaches to some extent. It enhances the intelligent reuse of existing process design cases and rule knowledge by facilitating the rapid generation of solutions, thereby facilitating the swift accumulation of process solution knowledge.

2. Related work

2.1. *Smart automated computer-aided process design*

Process design constitutes a crucial decision layer between product design and manufacturing production. It specifies the required process paths, equipment fixtures, and processing conditions, facilitating the transformation of designed components from raw materials into actual physical parts effectively and economically. The quality of the product, the efficiency of the manufacturing process, and the ultimate production cost are significantly influenced by the effectiveness of process planning. Traditionally, process planning relied on human expertise, and due to a lack of automation and intelligence, decision efficiency was limited, and optimal process planning could not be formulated (Gao, Mou, and Peng 2016). The rapid development of information technology has gradually

replaced manual processes in traditional process design with CAPP. CAPP generates process solutions from design models as a bridge between CAD and CAM. Process planning in CAPP should adhere to design specifications and manufacturing practices, accounting for complex relationships between operations and the uncertain state of manufacturing resources. This complexity renders it a dynamic decision problem, with the primary challenge lying in the acquisition, representation, and utilisation of process knowledge derived from human experience (Yusof and Latif 2014). Generally, CAPP can be categorised into two basic types: retrieval-based design and generative design. While some manufacturing enterprises still utilise retrieval-based approaches to assist in process design, generative design is emerging as a crucial trend in the development of intelligent process design. As more researchers study automated and intelligent CAPP systems, various technologies and methods have emerged, including feature-based methods and artificial intelligence methods (Besharati-Foumani, Lohtander, and Varis 2019).

Various feature-based process reuse techniques have been proposed and widely applied in the manufacturing industry. Process templates constructed based on similar features, machining operation sequences, process parameters, and machining principles are stored in a knowledge repository for reuse, aiming to reduce unnecessary user interactions (Huang et al. 2019b). Jong et al. (2015) introduced an automated process planning method for mold components that integrates feature recognition and grouping techniques. However, this method groups similar parts using part codes, and the granularity of process reuse is relatively broad, falling into the category of part-level reuse for machining process solutions. Zheng et al. (2008) proposed a system modelling and reuse method based on multi-granularity process knowledge. However, retrieval accuracy is compromised due to the inadequacy of group technology in fully describing part information. In summary, feature-based methods offer a dependable reservoir of feature knowledge, facilitating the reutilisation of process knowledge and enhancing the efficacy of process programme design. However, these methods are restricted to matching the reuse of pre-existing knowledge, thereby lacking the capacity for reasoning to extract process knowledge and encountering issues with inflexible reutilisation.

To address the limitations of the methods above, an artificial intelligence approach has been proposed. Traditional research has focused on heuristic optimisation methods, defining process planning tasks as multi-constraint combinatorial optimisation problems. Vafadar, Hayward, and Tolouei-Rad (2017) proposed a simulation model that utilises genetic algorithms to find optimal process parameters and tool configurations for dedicated machines. Although optimisation algorithms can effectively find approximate solutions, the search process is time-consuming, thereby necessitating enhancements in terms of flexibility and reusability. As an alternative to heuristic optimisation algorithms, Deep Reinforcement Learning (DRL) holds potential for addressing complex and dynamic decision problems. For instance, Wu et al. (2021) employed DRL for rapid decision-making in process planning using dynamic machining resources. Furthermore, early Knowledge-Based Systems (KBS) were also considered crucial to assist in process design. Li, Xie, and Xu (2011) provided a review of the development of KBS, introducing knowledge acquisition techniques, rule-based and case-based system approaches for developing knowledge-based software tools, and offering insights into future research directions and the real-time application of knowledge in manufacturing. Kuegler et al. (2023) conducted a systematic review of Knowledge-Based Engineering (KBE), emphasising KBE's temporal and financial

benefits in terms of knowledge reutilisation and design automation. They elucidated the deficiencies in current methodologies and techniques to resolve intricate professional engineering issues.

In conclusion, the methodologies outlined above demonstrate applicability within a defined scope of process design scenarios; however, they cannot currently address the demands of more intricate process solution design generation, particularly regarding process knowledge reuse and reasoning capabilities. Consequently, it becomes imperative to integrate more sophisticated and intelligent technologies and methodologies to tackle the deficiencies observed in traditional CAPP methods concerning the adaptable reuse of process knowledge and the formulation of reasoned process solutions. This entails leveraging advanced techniques, such as amalgamating the strengths of knowledge graphs for process knowledge representation and solution reasoning.

2.2. Knowledge graph in machining process design

In recent years, artificial intelligence technology has been extensively researched and applied in industrial manufacturing, showcasing solid capabilities in handling large-scale data and modelling complex mapping relationships (Zeba et al. 2021). Its development has injected new momentum into the manufacturing industry, driving a transition from automation to intelligence. Therefore, process design must evolve towards intelligence to meet intelligent manufacturing requirements. The widespread application of traditional KBS in process planning has demonstrated that more intelligent knowledge graphs are also suitable for process design (Xiao et al. 2023). Process planning heavily relies on process knowledge at every stage of process analysis and decision-making. For example, parts may have similar machining processes, and the reusability of process knowledge directly impacts the quality and efficiency of part design (Wen et al. 2022). The accurate representation and utilisation of process knowledge are crucial for developing intelligent process planning systems. However, due to various bottlenecks in the representation, matching, and reasoning of machining process knowledge, process planning still significantly relies on the knowledge and experience of process designers. Many researchers have applied KG to process planning, and using KG technology to expand the manufacturing domain has become a definite trend, representing a vital aspect of the transformation from traditional manufacturing to intelligent manufacturing (Yan, Yang, and Wan 2020). Substantial scientific achievements have been obtained in this context.

In process knowledge modelling, researchers have explored various methods for representing process knowledge to support applications. Li et al. (2018) comprehensively reviewed knowledge representation methods in mechanical machining process planning. They discussed 698 papers on knowledge representation methods, introducing applications in feature recognition, tool selection, fixture planning, operation selection and sequencing, and Computer Numerical Control (CNC) machining solution generation. Zhang et al. (2014) proposed an ontology-based knowledge representation model for feature-based machining operations (turning or milling), thereby constructing a machining knowledge repository. Eum et al. (2013) presented an ontology-based process planning knowledge modelling method. This method utilises relevant attributes to match knowledge about features, machining methods, and capabilities. It enables knowledge-based selection of corresponding machining methods for multi-axis machining features. Sormaz and

Sarkar (2019) constructed a manufacturing process planning knowledge repository model using a new foundational ontology (upper-level). This model facilitates interoperability with other upper-level manufacturing ontologies and the process planning design of prism components in a cloud manufacturing environment. Li et al. (2019) aimed to address the difficulty of reusing and sharing process knowledge. They used a knowledge graph to represent process knowledge uniformly and constructed a knowledge repository for mechanical machining. Wang et al. (2021) introduced a generalised solution representation model for process design, known as the Concept Space - Decision Space - Knowledge Space (CDK) Model. This model aims to map the solution design process into concept, decision, and knowledge spaces. They verified the feasibility and validity of this model, laying the groundwork for unifying case and rule knowledge in representing machining process solutions.

Researchers have combined various reasoning algorithms with knowledge graph to achieve novel processes in reusing process knowledge through inference. Zhang et al. (2022) integrated knowledge graphs with deep learning for macro-level process planning to enhance process generation efficiency. Dong et al. (2022) processed the process knowledge graph using case-based reasoning to retrieve and reuse existing knowledge, generating candidate process knowledge. Zhou et al. (2021) constructed a causal event evolution graph based on KG and tabular information. Subsequently, they employed a novel entity alignment method using neighbourhood sample graph convolutional networks to merge cross-domain knowledge graphs into a unified knowledge repository. (Wang et al. 2022) utilised matching algorithms to construct an intelligent question-answering system based on a turbine CNC machining knowledge graph employed for turbine CNC machining process generation. However, the aforementioned deep learning methods rely on data annotation, resulting in high data acquisition and model training costs. Furthermore, they struggle to address the requirements of highly specialised process design. The case-based reasoning method exhibits poor migration ability, making it unable to adapt to the need for a flexible, intelligent generation of process solutions. Additionally, traditional knowledge recommendation and knowledge quiz retrieval and matching methods face challenges in effectively mining and utilising process knowledge within KG, limiting them to only achieving essential knowledge reuse. KG's structured and interpretable characteristics, coupled with the robust environment-aware capability of deep reinforcement learning, offer a promising solution to these issues. Song, Zhao, and Wang (2021) represented KG as triplet vector spaces and employed them to construct a DRL model, thereby enhancing the efficiency and accuracy of rule mining in the model. Zhang et al. (2023) regarded knowledge graphs as explicit knowledge to improve the interpretability and training efficiency of deep reinforcement learning, validating the feasibility of combining KG and DRL for inference tasks. Despite several studies integrating DRL with manufacturing scenarios (Li et al. 2023), research on the application of KG and DRL in generating process design solutions remains relatively scarce.

In conclusion, drawing from prior research, this study delineates the attainment of unified knowledge representation and the intelligent reapplication of reasoning as primary objectives in the thoughtful formulation of machining process solutions. It introduces a machining process knowledge representation approach integrating the CDK model and proposes a solution generation framework based on DRL. This offers an efficient means for artisans to derive process solutions meeting design requisites from historical cases and general rules. Moreover, it addresses the constraints inherent in conventional methods,

precisely the rigidity in case knowledge reuse and the fragmented nature of rule knowledge reasoning. Ultimately, this endeavour fosters the swift accumulation of knowledge within machining technology.

3. Proposed method

3.1. Machining process knowledge representation integrated with CDK model

The manufacturing industry possesses abundant knowledge of the machining process, covering various aspects such as parts, operations, steps, features, and resources. To achieve a unified organisation and expression of different types of knowledge and to provide a knowledge representation foundation for knowledge-based machining process solution generation, this paper proposes a machining process knowledge modelling based on the CDK Model. In Figure 1, relevant concepts in the field of the machining process are distilled through research, resulting in a top-level knowledge model representing the fundamental structure and content of machining process design solutions. Secondly, based on the characteristics of knowledge accumulation in process design, the top-level knowledge model

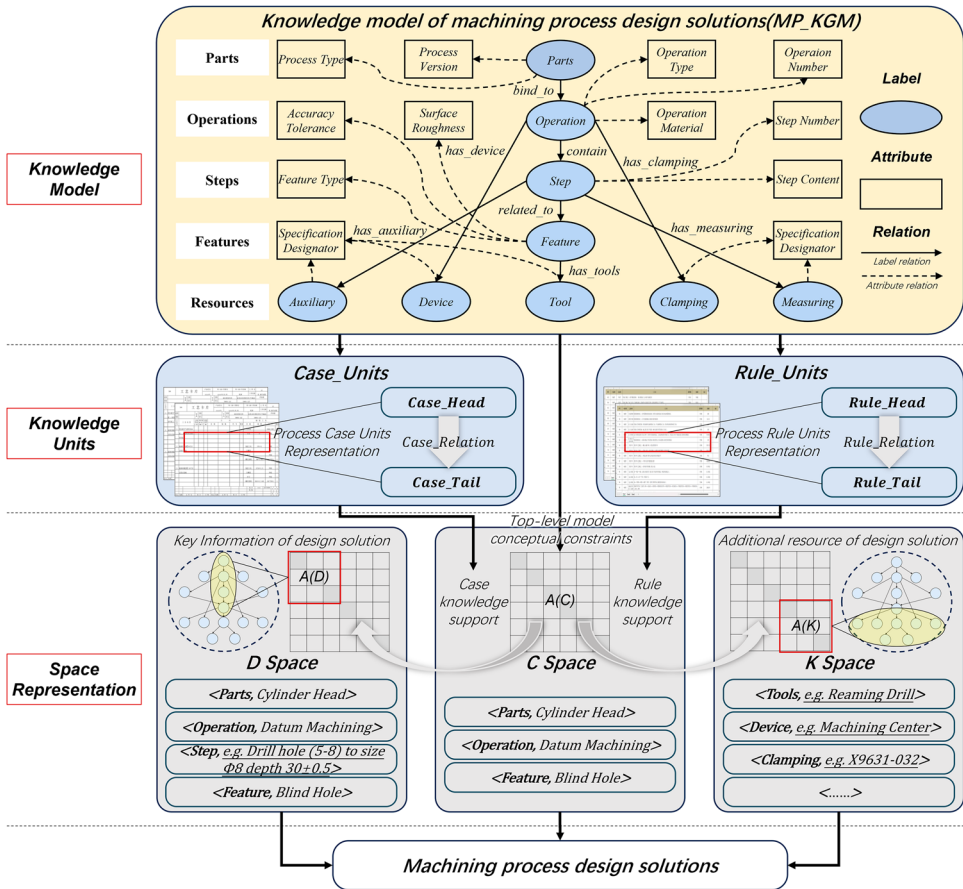


Figure 1. Machining process knowledge representation integrated with the CDK model.

for the machining process is divided into two categories: process cases with relatively complete knowledge structures and process rules with more scattered knowledge storage. These are uniformly represented as knowledge units in **(head, relation, tail)**. Finally, the machining process CDK space is defined, enabling the top-level knowledge model, case knowledge units, and rule knowledge units related to process design to be computed in adjacency matrix form. It lays the foundation for knowledge representation in generating solutions from requirements.

(1) **Knowledge model.** The pertinent entities and relational concepts within the domain of the machining process undergo refinement to derive the overarching knowledge model for machining process design. This model standardises the structure of the machining process knowledge graph, thereby facilitating the representation of case and rule knowledge. Additionally, the model delineates the boundaries of demand, solution structure, solution content, and other elements integral to the generation of machining process solutions, serving as the cornerstone of this paper.

(2) **Knowledge units.** Considering the characteristics of data resources and knowledge reuse within the process design domain, process knowledge is categorised into two distinct types: process cases embodying a comprehensive knowledge structure and process rules characterised by fragmented knowledge storage. Unified under the format of **(head, relation, tail)**, these are conceptualised as knowledge units. This framework aligns with the specifications of the top-level knowledge model for machining technology, enabling the construction of a comprehensive map of case and rule machining process knowledge. This facilitates easy retrieval and matching of process case and rule knowledge to generate process solutions.

(3) **Space representation.** The process of generating machining process solutions is deconstructed into three sequential steps: initial requirement determination, essential information selection, and related resource supplementation. This facilitates delineating the machining process CDK space, wherein design requirements, process knowledge, and process solutions pertinent to machining process design are represented using adjacency matrices. This lays the groundwork for knowledge representation in solution generation from requirements and establishes the foundational framework for the amalgamation of KG and DRL in the generation of machining process solutions.

3.1.1. Top-level knowledge model in the machining process

By surveying relevant concepts in the field of machining process design, this section summarises and organises common process concepts in the machining process design process. The result is the top-level knowledge model *MP_KGM*, representing the fundamental organisational structure of a complete machining process design solution. This model is a crucial guiding factor for the expected goals in the solution-generation process. Additionally, this model is a prerequisite for categorising machining process knowledge into two main classes: cases and rules.

In this paper, starting from different hierarchical levels constituting a machining process solution, the top-level knowledge model is divided into five layers: Parts, Operations, Steps, Features, and Resources. These divisions comprehensively represent the various components involved in a machining process solution.

$$MP_KGM = \{P, O, S, F, R\}$$

Table 1. Entity definitions for the MP_KGM.

Layers	Labels	Attributes
<i>Parts Layer</i>	Parts	Part Name, Part Code, Part Drawing Number, Process Type, Process Category, Process Version
<i>Operations Layer</i>	Operations	Operation Name, Operation Number, Operation Type, Workpiece Material, Programme Number
<i>Steps Layer</i>	Steps	Step Name, Step Content, Step Number
<i>Features Layer</i>	Features	Feature Name, Feature Type, Dimensional Accuracy, Geometric Tolerance, Surface Roughness
<i>Resources Layer</i>	Auxiliary	Auxiliary Name, Auxiliary Code, Auxiliary Specifications
	Device	Device Name, Device Code, Device Specifications
	Tools	Tool Name, Tool Code, Tool Specifications, Tool Module, Module Code, Module Specifications
	Clamping	Clamping Name, Clamping Code, Clamping Specifications
	Measuring	Measuring Name, Measuring Code, Measuring Specifications

Table 2. Relationship Definitions for the MP_KGM.

<i>Labels</i>	<i>Parts</i>			<i>Operations</i>		<i>Steps</i>		<i>Features</i>		<i>Auxiliary</i>
<i>Symbol</i>	①			②		③		④		⑤
<i>Labels</i>	<i>Device</i>			<i>Tools</i>		<i>Clamping</i>		<i>Measuring</i>		...
<i>Symbol</i>	⑥			⑦		⑧		⑨		...
	①	②	③	④	⑤	⑥	⑦	⑧	⑨	
①		bind_to								
②			contain			has_device		has_clamping		
③				related_to	has_auxiliary					has_measuring
④							has_tools			
⑤										
⑥										
⑦										
⑧										
⑨										

Each level represents a category of elements in the machining process design, and each element is composed of corresponding labels, names, and attributes. The definitions of machining process entities with different labels at each level are shown in Table 1.

The relationships between entities in the obtained top-level knowledge model must also be defined for better comprehensibility, enhancing the model’s readability. The relationship definitions are presented in Table 2.

3.1.2. Machining process case and rule knowledge units

Considering that the current knowledge base for the machining process often stores machining process knowledge in either complete cases or scattered entries, it is challenging during the process design to effectively leverage the support from both types of knowledge for generating solutions. Therefore, this paper uses the knowledge related to process

cases and process rules to generate machining process design solutions. Both types' original, diverse knowledge structure knowledge structures are unified and decomposed into knowledge units based on the **(head, relation, tail)** triplet structure, where **relation = head – tail**. This restructuring facilitates subsequent adjacency matrix operations in the CDK space.

$$MP_KGM = \{Case_Units, Rule_Units\}$$

(1) Case knowledge units

$$Case_Units = \langle Case_Head, Case_Relation, Case_Tail \rangle$$

The complete machining process case knowledge is decomposed into triplets consisting of two nodes and their interconnected relationships To facilitate unified representation. It is defined as a *Case_Units* comprising *CH*, *CR*, and *CT*. For instance, *< Parts: Cylinder Head, bind_to, Operations: Machining of reference surface >* represents the allowance of performing datum surface machining on the Cylinder Head part, forming interdependencies and supportive relationships between the corresponding part and operation. The *Case_Units* is the fundamental unit based on the top-level knowledge model in the machining process, including the basic building block of a complete machining process case. It also serves as one of the knowledge bases for the inference and generation of machining process design solutions. As shown in Table 3, some case units are listed.

(2) Rule knowledge units

$$Rule_Units = \langle Rule_Head, Rule_Relation, Rule_Tail \rangle$$

Similarly, the scattered machining process rule knowledge is represented as individual triplets composed of two nodes and their connecting relationship, defined as rule knowledge units (*Rule_Units*) consisting of *RH*, *RR*, and *RT*. For instance, *< Features: Blind Hole, has_tools, Tools: Alloy Drill >* expresses that for the blind hole feature, an alloy drill tool is commonly chosen, with no quantified rule conditions between the corresponding feature and tool, only conventional rule descriptions. Rule knowledge units are the fundamental components for composing disparate entries in machining process rules based on the top-level knowledge model. They also constitute one of the knowledge bases for the inference

Table 3. Example of machining process *Case_Units*.

No.	Case_Head	Case_Relation	Case_Tail
1	Parts: Cylinder head	bind_to	Operations: Machining of reference surface
2	Parts: Upper crankcase	bind_to	Operations: Rough machining of mating surface
3	Operations: Machining of reference surface	contain	Steps: Leave 1 mm allowance on the upper surface
4	Steps: Leave 1 mm allowance on the upper surface	related_to	Features: Table surface
5	Operations: Machining of injector face and inclined hole	has_device	Device: Machining centre

Table 4. Example of machining process R_{lue}_Units.

No.	Rule_Head	Rule_Relation	Rule_Tail
1	Features: Blind holes with a surface roughness requirement of Ra3.2	has_tools	Tools: Drilling can be performed using tools such as expansion drills, alloy drills, and boring tools.
2	Operations: Hole processing route	contain	Steps: drilling, expanding, reaming, and hand reaming, with the most widespread application.
3	Steps: Milling-type processing	has_tools	Tools: Ball cutters and circular cutters are commonly used for machining surfaces and contours with varying angles.

and generation of machining process design solutions. As shown in Table 4, some rule units are listed.

3.1.3. Machining process CDK space representation

The above section has outlined the top-level models pertinent to machining process design, encompassing their case and rule knowledge units, thereby establishing a theoretical framework for the structural norms and content support integral to machining process design solutions. However, for the reinforcement learning model employed in this paper to comprehend and compute process solutions' structural and content information, a more detailed representation of the machining process is essential. In conjunction with the CDK space model and the decomposition principles of the problem-solving process, the generation process of machining process design solutions is delineated into three sequential steps: (1) determining initial machining process design requirements; (2) discerning essential information for the design solution; (3) supplementing the selection of relevant machining process resources. The connotations of the machining process C, D, and K spaces are explicated, representing each space as an adjacency matrix. This approach facilitates the unified calculation of process solutions' structural and content information through adjacency matrices, thereby establishing the groundwork for a knowledge-driven reinforcement learning solution generation.

(1) Machining process C space

① **Domain connotation.** The processing requirements space for typical part machining represents a diverse set of design requirements, denoted as C_{nc} , in the machining process design. It encompasses labels and attributes of machining process design requirement entities such as parts, operations, steps, features, resources, etc. The definition is as follows:

$$C_{nc} = \{Label_C_{nc}, Attribute_C_{nc}\}$$

$$Label_C_{nc}, Attribute_C_{nc} \in MP_KGM$$

Before formulating a machining art design solution, it is imperative to articulate specific process design requirements. These may include identifying the part to be machined, delineating existing tooling resource constraints, and specifying the targeted process or work step for emphasis in the generation process. To facilitate the alignment of essential information within the intricate and varied design requisites with the top-level knowledge model and to streamline the computationally intensive process of generating solutions based on case- and rule-based knowledge, this paper maps various types of machining process

design requirements proposed by process personnel into a unified requirement matrix $\mathbf{C}_M$ according to the structure of the top-level knowledge model, consisting of node table $\mathbf{N}(C)$ and adjacency matrix $\mathbf{A}(C)$, forming a process requirement space C that conforms to the top-level specification of machining process knowledge models and covers concepts of multi-level machining process design requirements.

② **Space representation.** The node table $\mathbf{N}(C)$ contains instantiated and uninstantiated knowledge entities or nodes composed of labels and attributes. The size of the C space is limited by the number of entity types in the top-level knowledge model, i.e. $nc \wedge 2$ order adjacency matrix $\mathbf{A}(C)$ with nc equal to the total number of entity label types N_{MP_KGM} in the top-level knowledge model. The $[0, 1]$ values in the adjacency matrix $\mathbf{A}(C)$ represent whether there is a connection between the starting entity corresponding to the row and the ending entity corresponding to the column, where 0 indicates no connection, and 1 indicates a connection.

(2) Machining process D space

① **Domain connotation.** The main decision space for the typical machining process of parts represents a crucial set of primary concepts, denoted as D_{nd} , that constitute the machining process design solution. The top-level knowledge model determines which concepts are vital entities based on the instance requirements concepts in the machining process C space. The D space comprises entities that form the minimal subgraph covering the machining process design requirements in the top-level knowledge model. The definition is as follows:

$$D_{nd} = \{Label_D_{nd}, Attribute_D_{nd}\}$$

$$Label_D_{nd}, Attribute_D_{nd} \in MP_KGM$$

Upon elucidating the primary machining design requisites, precedence should be given to process information closely aligned with the initial requirements. This includes pertinent details concerning components, features, tools, and other relevant factors. Subsequently, process concepts intricately linked to the information above will be integrated into the machining D -space, serving as pivotal entities for unified representation. To streamline the matching and computation of associated concepts, this study maps the requirements and related concepts proposed by process personnel as the backbone elements for solution generation into the decision matrix $\mathbf{D}_M$, consisting of node table $\mathbf{N}(D)$ and adjacency matrix $\mathbf{A}(D)$, forming a decision space D that conforms to the top-level specification of the machining process knowledge model and requires special attention in the solution generation process.

② **Space representation.** The node table $\mathbf{N}(D)$ contains instantiated and uninstantiated knowledge entities or nodes composed of labels and attributes. The size of the D space is determined by the number of nodes in the minimal subgraph of critical entities, i.e. nd^2 order adjacency matrix $\mathbf{A}(D)$ with nd equal to the number of entity label types $N_{mingraph}$ in the minimal subgraph of essential entities. It is less than or equal to the total number of entity label types N_{MP_KGM} in the top-level knowledge model. The $[0, 1]$ values in the adjacency matrix $\mathbf{A}(D)$ represent whether there is a connection between the starting entity

corresponding to the row and the ending entity corresponding to the column, where 0 indicates no connection, and 1 indicates a connection.

(3) Machining process K space

① **Domain connotation.** The supplementary resource space for the typical machining process of parts represents an additional set of resources, denoted as K_{nk} , that constitute the machining process design solution. Based on the instance requirements concepts in the machining process C space, where entities related to non-key entities in the decision space are considered supplementary resources, the K space is defined as follows:

$$K_{nk} = \{Label_K_{nk}, Attribute_K_{nk}\}$$

$$Label_K_{nk}, Attribute_K_{nk} \in MP_KGM$$

To generate a comprehensive machining art programme aligning with the constraints delineated by the top-level knowledge model, augmenting additional process information beyond the initial requirements is imperative. This augmentation is based on the primary information to enhance the programme outcomes and aid artisans in conducting thorough process design. Therefore, this paper uniformly maps relevant concepts, not explicitly stated by the artisans but specified by the top-level model, onto the resource matrix K_M as branching elements for solutions generation, consisting of the node table $N(K)$ and adjacency matrix $A(K)$, forming a supplementary resource space K that conforms to the top-level specification of the machining process knowledge model and plays an auxiliary role in the solution generation process.

② **Space representation.** The node table $N(K)$ includes instantiated and uninstantiated knowledge entities or nodes composed of labels and attributes. The size of the K space is determined by the number of nodes in the supplementary resources, i.e. $nk \wedge 2$ order adjacency matrix $A(K)$ with nk equal to the total number of entity label types N_{MP_KGM} in the top-level knowledge model minus the number of entity label types $N_{mingraph}$ in the minimal subgraph of critical entities, that is, $nk = nc - nd = N_{MP_KGM} - N_{mingraph}$. The $[0, 1]$ values in the adjacency matrix $A(K)$ represent whether there is a connection between the starting entity corresponding to the row and the ending entity corresponding to the column, where 0 indicates no connection, and 1 indicates a connection.

3.2. Deep reinforcement learning-based solution generation

By representing the machining process knowledge, the transformation of the machining process from design requirements to design solutions is turned into a computational process based on an adjacency matrix. Knowledge models and knowledge units provide spatial structure and content generation support for the reinforcement learning model. This section uses deep reinforcement learning models to generate machining process design solutions. Following requirement identification, model construction, and solution parsing, machining process design requirements are gradually expanded to design solutions that conform to the expected structure and satisfy existing process knowledge. Designers can efficiently obtain reference solutions that better align with the intent of machining process design, continually refine the proposed process requirements, and ultimately formulate

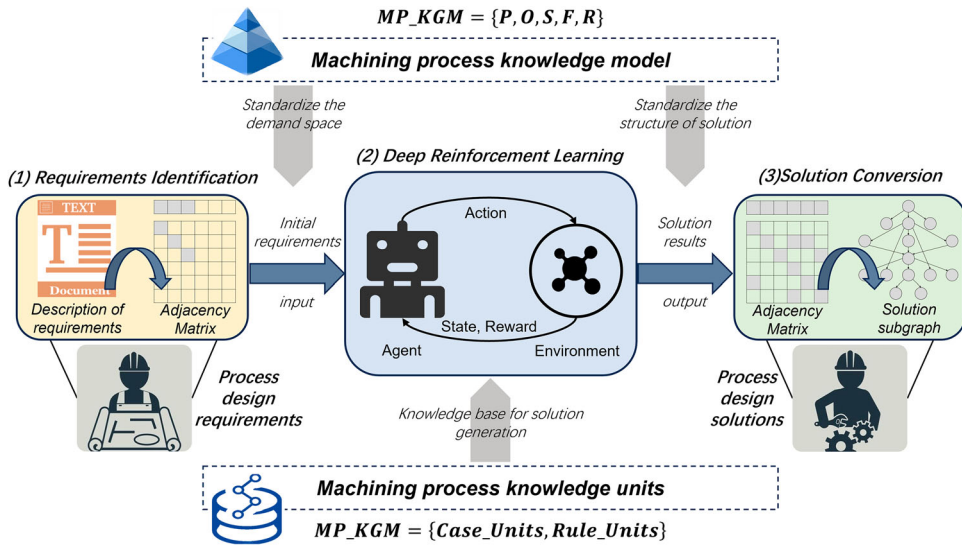


Figure 2. Framework for machining process solution generation using DRL.

new solutions suitable for guiding actual production. Validated solutions can be directly stored in the case knowledge base for rapid accumulation. As shown in Figure 2, this mainly involves:

- (1) **Requirements Identification.** Matching machining process design requirements with corresponding nodes in the machining process knowledge graph and transforming them into a requirement feature adjacency matrix with a specific structure and connotation according to the definition of the CDK space. This results in an initial requirement state space for deep reinforcement learning.
- (2) **Deep Reinforcement Learning.** Combining the representation of the machining process CDK space model, configuring the states, actions, and rewards for deep reinforcement learning, and finally, selecting an appropriate deep reinforcement learning model to construct a framework for generating process solutions based on machining process cases and rule knowledge, obtaining the adjacency matrix of the solution.
- (3) **Solution Conversion.** Transforming the output state space solution results in reinforcement learning into a visualised knowledge graph structure. It allows process designers to query and browse detailed information about the generated solutions. It facilitates incorporating the generated solutions as process case knowledge into the existing machining process knowledge graph.

3.2.1. Identification of machining process requirements

To better integrate machining process design requirements into the reinforcement learning generation process of process solutions, it is essential to first adequately identify the structure and content of diverse process design requirements. It allows process requirements to match specific machining process knowledge graph instances. Subsequently,

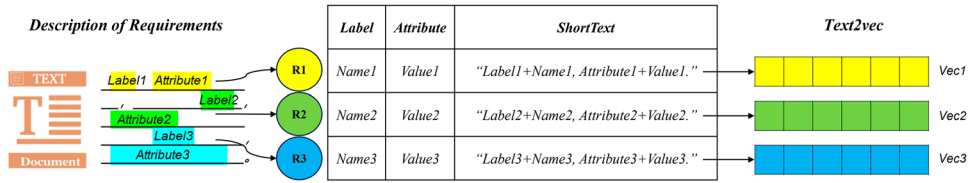


Figure 3. Extraction of process requirement nodes and text representation.

based on the definition of the CDK space model in Section 3.1.3, machining process elements related to process requirements are allocated to the corresponding space. Finally, a merged adjacency matrix is obtained, representing the features of process requirements and the structure of process solutions. This matrix establishes the initial requirement state for the knowledge-based deep reinforcement learning model's state space, action space, and reward mechanism.

(1) Representation of process requirements

Utilising the top-level knowledge model of the machining process, process personnel extract relevant process concepts from design requirement texts through natural language processing techniques such as segmentation and named entity recognition. This extraction forms a requirement list consisting of labels and attributes, transforming process text into the format of process nodes, as shown in Figure 3. To ensure that the extracted process nodes match relevant entities stored in the machining process knowledge graph, the labels and attribute information contained in the nodes *Label: Name, Attribute: Value* are combined into short texts. Text representation tools such as text2vec are used to calculate the similarity between process requirement nodes and knowledge graph nodes. The matching result with higher similarity is chosen as the initial instantiation representation for the process demand.

(2) Representation of process solution

Based on the definition of the machining process CDK space representation, the instantiation representation of the initial process requirement is integrated into the process requirement adjacency matrix, yielding the adjacency matrix $\mathbf{A}_0(C)$. Simultaneously, the process of generating machining process solutions is abstracted as the gradual addition of instantiated nodes and relationships to the adjacency matrix. Consequently, the features in $\mathbf{A}_0(C)$ need to be allocated to spaces D and K, representing critical information on machining process solutions and supplementary resources. This recombination results in an initial solution space representation suitable for reinforcement learning-based solution generation.

As shown in Figure 4, to transform the requirement space into the solution space, concepts in the C space need to be allocated to the D and K spaces. Following the instantiation of process requirements, the minimum subgraph covering the proposed process requirement concepts is selected under the top-level knowledge model specifications. The concepts within the minimum subgraph are assigned to the D space, resulting in the adjacency

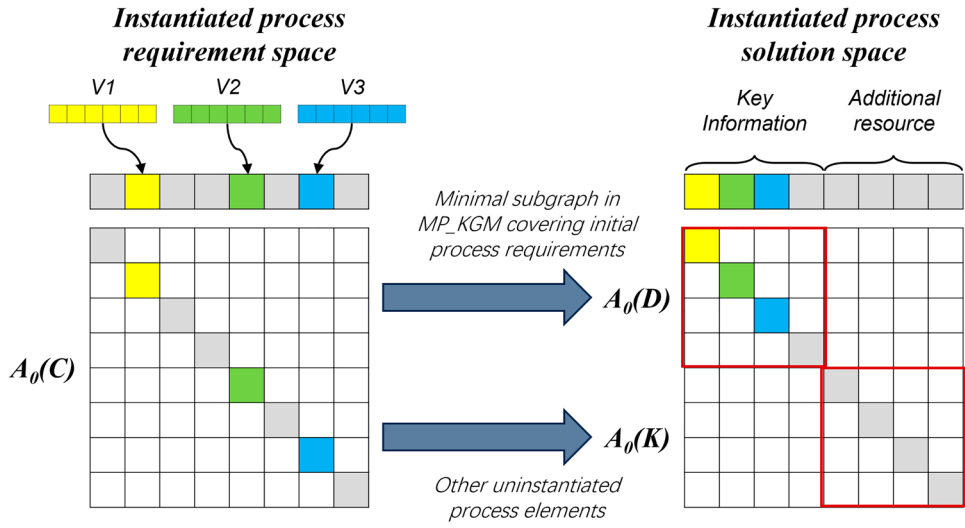


Figure 4. From requirement space to solution space.

matrix $\mathbf{A}_0(D)$ representing essential machining process information. Subsequently, the concepts not considered in the minimum subgraph are allocated to the K space, generating the adjacency matrix $\mathbf{A}_0(K)$ representing supplementary machining process resources. The specific process is outlined below:

- ① Using a combinatorial approach, select unordered pairs of demand labels and define them as $Start_C$ and End_C .
- ② Match in the knowledge model the undirected knowledge chain that satisfies the shortest length requirement with the chosen two demand labels as the start and end, denoted as $\langle Start_C, Label_1, \dots, Label_x, End_C \rangle$.
- ③ Merge all combination matches of knowledge chains, remove duplicate start, end, and intermediate labels, obtain the minimum knowledge subgraph covering the initial demand, and represent it as the adjacency matrix $\mathbf{A}_0(D)$. The matrix dimension is p , where the adjacency list is arranged with instantiated labels in the front and uninstantiated labels in the back.
- ④ Treat the remaining uninstantiated process labels as supplementary resources, represented as $\mathbf{A}_0(K)$, with a matrix dimension of q .
- ⑤ Merge $\mathbf{A}_0(D)$ and $\mathbf{A}_0(K)$ into one adjacency matrix $\mathbf{A}_0(D \& K)$, representing the initial solution space, denoted as $\mathbf{S}_0$. The upper-left $p \times p$ region of the matrix is the adjacency matrix for D space, the lower-right $q \times q$ region is the adjacency matrix for K space, and the values in the remaining region are defaulted to 0.
- ⑥ The resulting solution space $\mathbf{S}_0$ only contains instantiated representations of initial demands. Other entities and relationships await instantiation generation by the reinforcement learning model. Therefore, $\mathbf{S}_0$ serves as input to the reinforcement learning model, participating in model training and testing.

3.2.2. Deep reinforcement learning model

By precisely identifying machining process requirements, we seamlessly integrate these process design demands into the machining process's adjacency matrix. It provides the

fundamental structure and initial content for reinforcement learning in solution generation. It establishes a solid foundation for the involvement of triple-structured case and rule knowledge units in machining process solution generation. While formulating the machining process solutions using machining cases and rule knowledge, it is imperative to acknowledge that the amalgamation of knowledge units within the ternary structure cannot be executed simplistically. Additionally, the interdependence of each knowledge node within the process cases mandates that they cannot be disassociated from the entire case and must subsist cohesively. Furthermore, selecting appropriate rule knowledge becomes crucial when existing cases fail to meet the requirements. In this way, we can develop a machining solution generation model adept at seamlessly integrating machining cases and rule knowledge, thus catering to the needs of flexible process design.

Against this backdrop, the proposed solution generation based on deep reinforcement learning, as illustrated, will consider the conditions mentioned above extensively when setting the state space, action space, and reward mechanism. It ensures that the generated machining process solutions meet process design requirements and consider the effective involvement of triple-structured units. The state space encapsulates the content and configuration of the machining solution throughout the generation process, reflecting the synthesis of case and rule knowledge. In contrast, the action space denotes the specific execution involved in updating the state, adding new case and rule knowledge triads within the state space. Moreover, the reward mechanism delineates the incentive derived from each action execution, guiding the model to utilise case and rule knowledge for process solution generation adeptly. Within this framework, the deep reinforcement learning model progressively crafts machining process design solutions comprising case and rule knowledge through continuous interaction with the machining process knowledge graph. It learns to leverage the adaptability of rule knowledge while upholding the integrity of case knowledge in the machining process solution generation paradigm under the tutelage of the reward mechanism and continual training iterations.

Figure 5 (a) depicts a deep reinforcement learning model based on the Deep Deterministic Policy Gradient (DDPG) framework for machining process solution generation. It employs deep neural networks as approximations for the policy network and action-value function, trains the parameters of the policy network and value network models using stochastic gradient descent, introduces a twin neural network for a more stable learning process, accelerates the convergence process, and incorporates an experience replay mechanism to eliminate sample correlations and dependencies, making the algorithm more prone to convergence.

The detailed introduction of the state space, action space, and reward mechanism in the algorithm flow is as follows:

(1) State space

This paper's state space represents the structural and content states of the machining process design solutions. The structural form of a solution is defined by the top-level knowledge model of the machining process, representing conceptual elements (e.g. parts, operations, steps, features, etc.) and their interrelationships (e.g. binding, containment, involvement, etc.), using uninstantiated knowledge models to characterise the structure of the solution. Machining process cases and rule-based knowledge define the content state

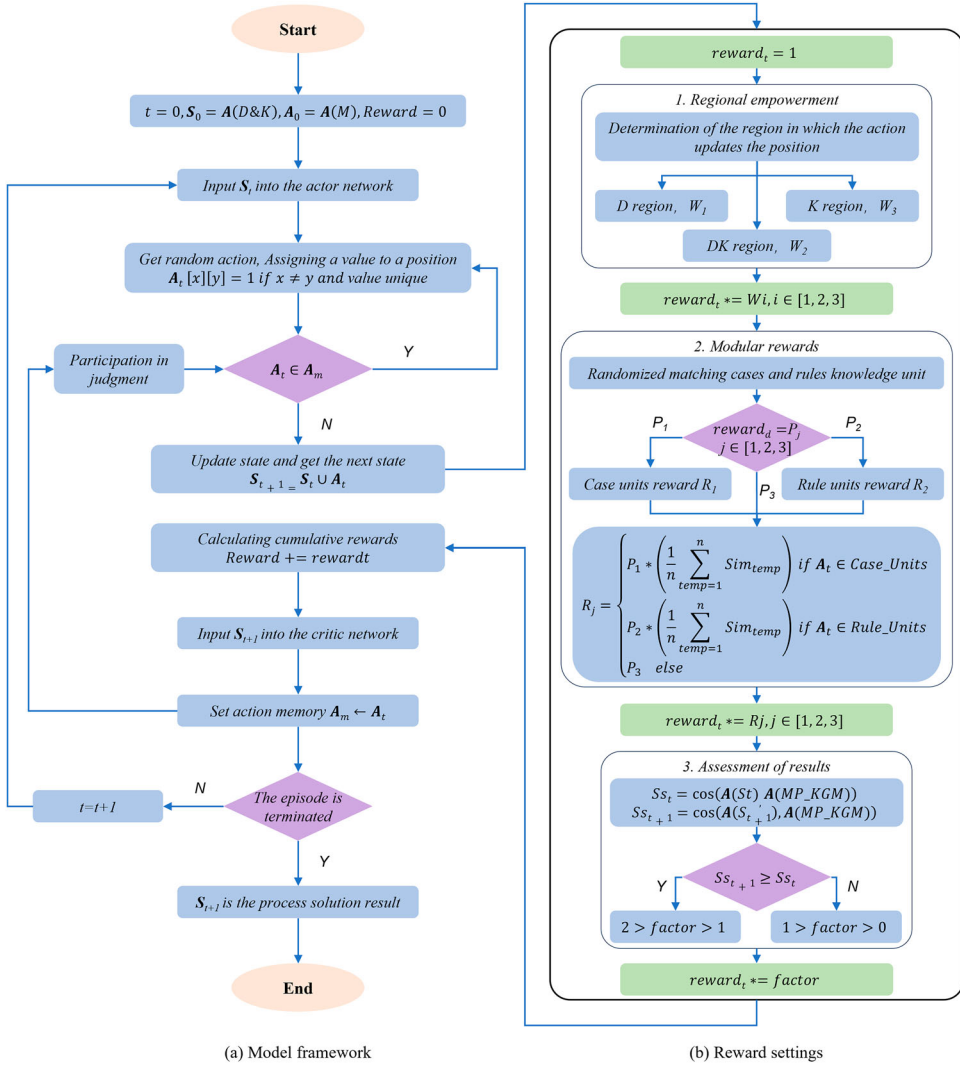


Figure 5. Algorithm flow.

of a solution. During the solution generation process, the reinforcement learning model gradually refines the content of the solution by acquiring case or rule-based knowledge from the machining process knowledge graph. To facilitate model computation and learning, the structural and content states of the solution are represented as a state matrix $\mathbf{S}$, with the structure and content of the solution represented separately using an adjacency matrix and a node table. The dimension of the adjacency matrix is the square of the sum of the number of nodes in the top-level machining process knowledge model plus one, i.e. $S_dim = (N_{MP_KGM} + 1)^2$.

As the generation of machining process solutions originates from design requirements, the initial state space $\mathbf{S}_0$ input to the model is the initial solution space $\mathbf{A}(D\&K)$, obtained through machining process requirement identification, as detailed in Section 3.2.1. Based on the initial state space, the solution generation process involves updating the state space.

Each instantiation of a node vector in the node table of the state matrix represents the addition of a machining process entity to the solution. In contrast, each change from 0 to 1 in a non-diagonal position of the adjacency matrix indicates creating a new machining process relationship. After the state space is updated, the matrix result $\mathbf{S}_{t+1}$ output by the model represents a machining process design solution.

(2) Action space

For the definition of the corresponding state space, the task of the action space is to update the structure and content of machining process solutions using stored case and rule knowledge. Combining the definition of the state matrix, we transform the process of adding machining process cases and rule knowledge units with the **(head, relation, tail)** triplet structure to the state space into operations of instance selection for head or tail entities in the node table and relationship creation at non-diagonal positions in the adjacency matrix. To facilitate computation, each action is represented as an action matrix $\mathbf{A}$. The vector representations of the head entity and tail entity, along with their label index, are essential information for executing actions, forming a simplified matrix of dimensions $A_dim = 2 \times 2$, i.e. $[[\mathbf{Vector}_h, Index_h], [\mathbf{Vector}_t, Index_t]]$. It significantly reduces the dimensionality of the action space while not affecting the ability of the action matrix $\mathbf{A}$ to update the state matrix $\mathbf{S}$. The steps executed by the action space $\mathbf{A}_t$ in each action are as follows:

- ① $t = 0$;
- ② From the initial state space $\mathbf{S}_0$, randomly select a processing entity instantiated representation as the head entity head from the node table and note its index position as a row;
- ③ Establish a new relationship for the first time, starting from the position $[row, row]$ in the initial state space $\mathbf{S}_0$, randomly change a value of 0 to 1 at a non-start point in the same row, creating a relation departing from the head, and denote the index of the new relation as $[row, col]$;
- ④ Based on the index of the relation, select the process entity as the tail from the node table at the index position col and determine whether the tail entity is instantiated. If not instantiated (value is 0), randomly match an entity vector from the machining process knowledge graph corresponding to a case or rule knowledge; otherwise, maintain the original instantiated vector representation of the machining knowledge;
- ⑤ With the above steps, obtain a triplet where both head and tail entities are instantiated, and a relationship is established between them. Then, derive the action matrix $\mathbf{A}_t$ corresponding to the indexes of the elements in the node table, adjacency matrix, and triplets. Compute $\mathbf{S}_t \cup \mathbf{A}_t$, replace positions with different values using the values of $\mathbf{A}_t$, and obtain the updated state space $\mathbf{S}_{t+1}$;
- ⑥ Store each executed action space $\mathbf{A}_t$ in the action memory space $\mathbf{A}_m$ for checking action repetition;
- ⑦ When $\mathbf{A}_m \neq \emptyset$, in step ②, establishing a new relationship should be based on the action memory space $\mathbf{A}_m$, and no relationship should be repeated in positions where relationships have already been established;
- ⑧ $t = t + 1$, loop through the above steps until the termination condition is met.

The pseudo-code for the action space is shown in Algorithm 1.

Algorithm 1 Get action of action space**Input:** S_t solution state space, *MPKG* machining process knowledge graph**Output:** A_t Simplified matrix containing information on required execution actions

```

1.  $S_0$  //Initial state space;
2.  $A_0$  //The initial action space defaults to  $\emptyset$ ;
3.  $A_m$  // Initialise to 0 matrix;
4. for ( $t = 0$  to  $maxstep$ ) do
    // Set the maximum step size and loop through the action
5.    $A_m = \text{Merge all } A_t$ ;
6.   for each  $head$  in  $S_t$  do
    // Randomly obtain instantiated entities  $head$  from the state space
7.      $Index_h = row$ ; // Index position of the record header entity
8.      $Vector_h = S_t[0, Index_h][vector]$ ; // Instance vector assignment
9.     for each  $tail$  in  $S_t$  do
10.       $Index_t = col$ ;
11.       $relation = [row, col]$ ;
12.      if  $A_m[relation] \neq 1$  and  $S_t[relation] \neq 1$  and  $Index_h \neq Index_t$  and  $S_t[0, Index_t][vector] = 0$  then
13.         $Vector_t = MPKG[S_t[0, Index_t][label]]$ ;
        //Randomly selected entities whose labels in MPKG match the label of the  $tail$  index position in  $S_t$  are
        assigned values
14.      end if
15.    end for
16.  end for
17.  return  $A_t = [[Vector_h, Index_h], [Vector_t, Index_t]]$ ;
18. end for

```

(3) Reward mechanism

The state space and action space have delineated the environment and procedures essential for generating machining process solutions. However, directing the model to adeptly integrate the case and rule knowledge stored within the machining process knowledge graph necessitates the establishment of a reward mechanism aligned with the machining process solution design process. This mechanism should seamlessly amalgamate knowledge from various structures. To address this, we devised a reward mechanism for machining process solution generation comprising regional weights, unit rewards, and result evaluation. As shown in the flowchart of Figure 5 (b), The nuances and importance of each component are elucidated below.

① Regional weight

According to the definition in section 3.2.1, the state space obtained by merging the D and K feature spaces implicitly encompasses the emphasis and connotations of D and K in the design of machining process solutions. Thus, the state space can be divided into three regions: D, DK, and K, each region influencing rewards as follows:

The state update in the D region is based on the extension of instantiated representations of proposed process requirements and relationship exploration. To ensure that the generated solutions align closely with the design requirements of process personnel and to emphasise the utilisation of existing instance information, the rewards brought by the D region's state update need to be highlighted. Therefore, the reward weight W_1 is set for the D region.

The state update in the DK region represents the creation of nodes and relationships between critical entities of the machining process and supplementary resources. Since the state in the D region is crucial for the solution, and the creation of relationships in the closely

associated DK region should also be rewarded, the reward weight W_2 for the DK region is set.

The state update in the K region represents the creation of nodes and relationships among supplementary resources that are not closely associated with the initial process design requirements but are essential for a complete machining process design solution. Hence, the reward weight W_3 is set for the K region.

Thus, the following weight relationships are established: $W_1 > W_2 > W_3$. Based on the initial reward that $reward_t = 1$, each reward is weighted to calculate $reward_t = reward_t * W_i$, $i \in [1, 2, 3]$.

② Units Reward

To ensure that the reinforcement learning model fully leverages existing machining process knowledge to generate the required process solutions, it is necessary to define the reward direction and value for the process of obtaining rewards for matching case and rule unit knowledge.

Firstly, the reward direction is divided based on the newly added process knowledge after state updates. If the newly added knowledge is present in the current machining process cases and rule knowledge graph, i.e. the triple $\langle \text{head}, \text{relation}, \text{tail} \rangle$ corresponding to the action space $\mathbf{A}_t$ can find the corresponding case or rule knowledge unit (denoted as $\mathbf{A}_t \in \{\text{Case_Units}, \text{Rule_Units}\}$), then the reward for this instance is considered positive. Conversely, the reward is considered harmful if the new knowledge does not exist in the knowledge graph. During calculation, the reward direction is denoted as $reward_d = P_j$, where $j \in [1, 2, 3]$. Here, P_1 represents the reward direction when the new knowledge belongs to the case knowledge unit and is greater than 0; P_2 represents the reward direction when the new knowledge belongs to the rule knowledge unit and is also greater than 0; P_3 represents the reward direction when the new knowledge does not belong to existing knowledge and is less than 0. Since case knowledge units are embedded within complete machining process cases, a robust structural correlation exists among them. Throughout the solution generation process, these case knowledge units serve as the primary constituents and as pivotal elements in ensuring the overall coherence of the generated solutions. Given their dispersed and adaptable nature, rule knowledge can be introduced and substituted within existing cases as per specific requirements. Consequently, this paper establishes a machining process solution generation framework with case knowledge as the primary component and rule knowledge as the supplementary element. Therefore P_j has the following relationship: $P_1 > P_2 > P_3$.

Next, combined with TransE for vector representation of knowledge triplets **head** + **relation** = **tail**, calculate the reward values R_j for the three reward directions P_j corresponding to each newly added knowledge in the action space $\mathbf{A}_t$. Given that each newly added knowledge in the action space can be represented as the structure $\langle \text{head}, \text{relation}, \text{tail} \rangle$, where **head** and **tail** are text representation vectors, we have **relation** = **tail** – **head**, abbreviated as $\mathbf{r} = \mathbf{t} - \mathbf{h}$. To reasonably set the reward values for each newly added knowledge, we use the case and rule knowledge units in the machining process knowledge graph with the same labels as the head and tail entities of the new knowledge as a basis. We represent knowledge units as $\mathbf{r}_U = \mathbf{t}_U - \mathbf{h}_U$, and calculate the cosine similarity $\cos(\mathbf{r}, \mathbf{r}_U)$ for each new knowledge, with each knowledge unit having the same label, denoted as Sim_n , where n represents the number of knowledge units with the same label. The average value of Sim_n represents the reward value for the new knowledge. Specifically, in the scenario

where the augmented knowledge corresponds to case knowledge and a Sim_n value of 1 is present, indicating a perfect match with an existing case, the mean value of Sim_n is set to the target value of 1.

This approach allows us to measure the relative importance of each newly added knowledge in the existing machining process knowledge. When the average similarity is more extensive, it indicates that the new knowledge aligns more with conventional machining process knowledge and, therefore, the reward value is more considerable. Conversely, when the average similarity is more minor, it indicates that the new knowledge tends to be more of an exception, so the reward value is smaller. Notably, for the P_3 case, where the new knowledge has no corresponding knowledge unit, the reward value R_3 equals P_3 . Combining region weights and reward directions, the definition of each reward is as follows:

$$reward_t = \begin{cases} reward_t * W_i * P_1 * \left(\frac{1}{n} \sum_{temp=1}^n Sim_{temp} \right) & \text{if } A_t \in Case_Units \\ reward_t * W_i * P_2 * \left(\frac{1}{n} \sum_{temp=1}^n Sim_{temp} \right) & \text{if } A_t \in Rule_Units \\ reward_t * W_i * P_3 & \text{else} \end{cases}$$

③ Result evaluation

To ensure that the state space of each generated machining process design adheres to the structural specifications of the top-level knowledge model, the cosine similarity between the adjacency matrix $\mathbf{A}(S_t)$ of the state space $\mathbf{S}_t$ at each step, and the adjacency matrix $\mathbf{A}(MP_KGM)$ corresponding to the top-level knowledge model is calculated. It is denoted as Ss_t And serves as a scaling factor for rewards. When the similarity between the state matrix and the top-level model increases, the reward value is amplified; conversely, when the similarity decreases, the reward value is diminished.

$$Ss_t = \cos(\mathbf{A}(S_t), \mathbf{A}(MP_KGM))$$

$$Ss_{t+1} = \cos(\mathbf{A}(S_{t+1}), \mathbf{A}(MP_KGM))$$

$$factor = \begin{cases} (1, 2) & \text{if } Ss_t \geq Ss_{t+1} \\ (0, 1) & \text{if } Ss_t < Ss_{t+1} \end{cases}$$

In summary, let $reward_t = reward_t * factor$, obtaining the reward value for each state update after executing an action and subsequently accumulating the total reward value.

3.2.3. Conversion of generated result solutions

When the model round terminates, the final state space $\mathbf{S}_{t+1}$ is output, which includes the node table and adjacency matrix. To enhance the readability of the output and facilitate the accumulation of machining process knowledge, the text vectors in the node table are parsed into specific machining entities in the knowledge graph. It is done by establishing the relational structure among machining entities based on the adjacency matrix. Additionally, labels are assigned to each relationship according to the definition of the top-level knowledge model. In the end, a knowledge graph-formatted machining

process design is obtained. It allows process engineers to retrieve detailed information about generated solutions through the knowledge graph, promoting easy data access for knowledge accumulation in the context of knowledge graph-based machining process knowledge.

4. Case study

To demonstrate the efficacy of the proposed methodology, this section employs an exemplar case involving the machining process design for a pivotal component of a specific model diesel engine. A machining process knowledge graph is constructed from pertinent data derived from process cards and rule entries tailored to diesel engines. After this, the system formulates machining process solutions aligned with the specified design criteria using a proficiently trained reinforcement learning model. This methodology serves as a benchmark, fostering the enablement of intelligent applications within machining process design. It achieves this by augmenting the organisation and reusability of intricate and specialised process knowledge. Furthermore, it substantively contributes to the progression of intelligent process design.

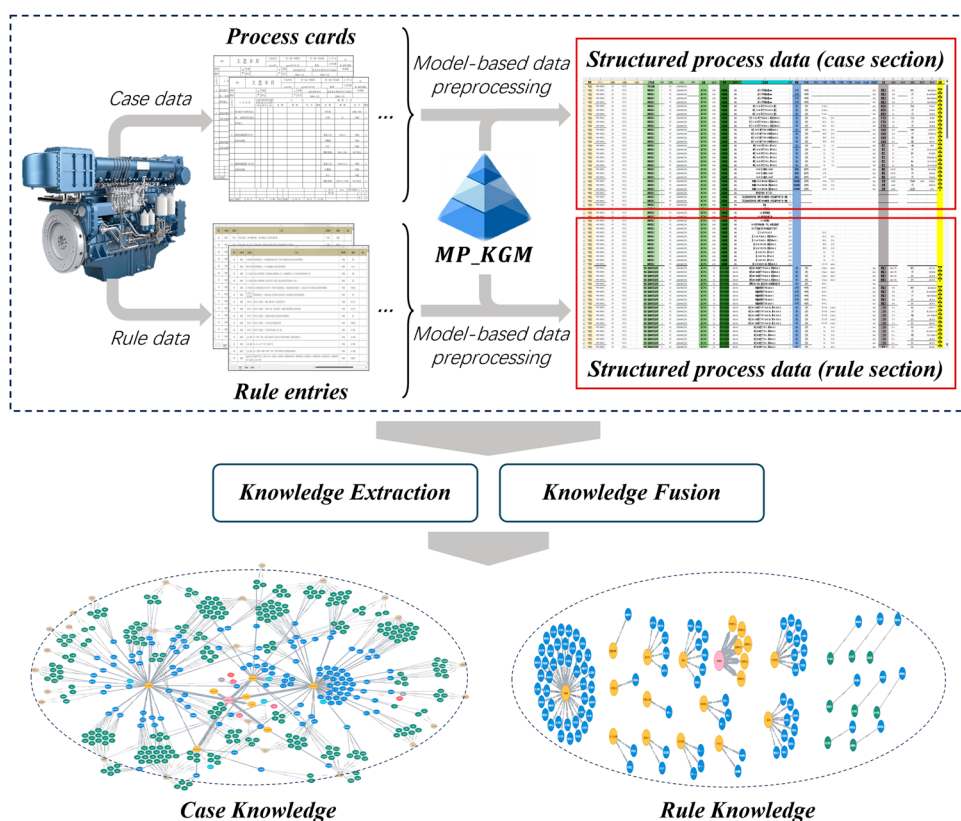


Figure 6. Construction of knowledge graph for diesel engine machining process.

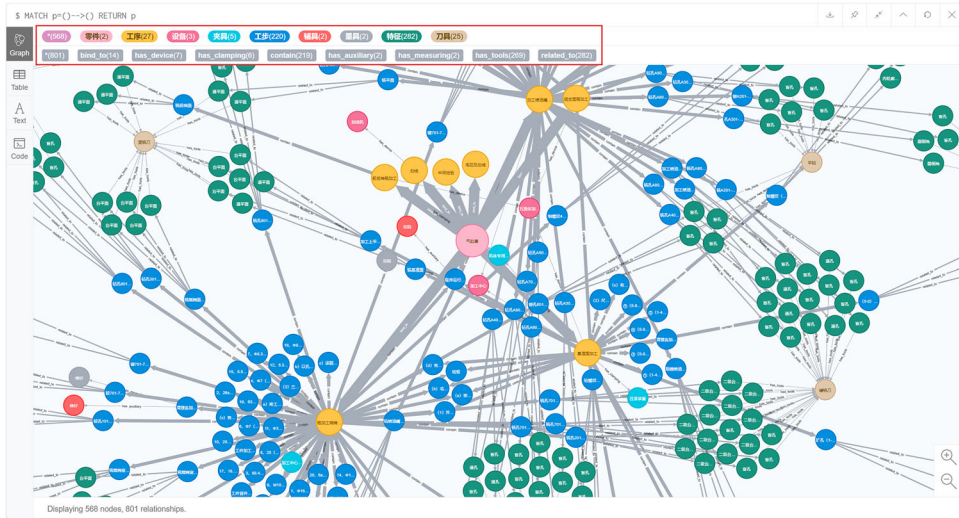


Figure 7. Machining process knowledge graph.

4.1. Knowledge preparation

The generation of machining process solutions relies on the support of the machining process knowledge graph. To rapidly obtain a knowledge graph containing machining process cases and rule knowledge, as shown in Figure 6, this paper preprocesses data from machining process cases and rule files representative of diesel engine machining, unifying relevant files into structured tabular data based on the definition of the top-level knowledge model for machining process. The required machining process knowledge graph is quickly constructed through mature knowledge graph techniques such as template-based knowledge extraction and semantic similarity-based knowledge fusion (Guo et al. 2024).

As shown in Figure 7, the resulting knowledge graph has 570 nodes and 801 relationships. Using the node count as a reference, the ratio of case knowledge to rule knowledge is approximately 4:1. Following the description in Section 3.1.2, combined with the diesel engine machining process knowledge graph, case knowledge forms knowledge subgraphs centred around components. For example, the machining sequence of the cylinder head component involves processes such as blanking and marking, machining of reference surfaces, and rough machining of combustion chamber surfaces and inclined surfaces. The machining of the reference surface consists of seven steps involving machining features such as table surfaces, through-holes, blind holes, and circular chamfers. It requires the participation of relevant resources such as machining centres, clamping devices, and alloy drills. These case knowledge instances collectively constitute a machining process case for the cylinder head, one of the critical components of the diesel engine. Rule knowledge, on the other hand, forms relatively independent and universally applicable knowledge chains. For instance, a clamping device with a specific code is used when processing flat features, indicating that this fixture has significant usage restrictions. The knowledge graph represents this by connecting the clamping device and flat feature nodes. From the visual representation of the graph, it is evident that case knowledge has more associated relationships and is more cohesive, providing a continuous knowledge source for generating

Table 5. The requirements for machining process design.

No.	Requirements
1	During the machining of the reference surface for the cylinder head component on the machining centre with the code TH6563 × 63A, how should the blind hole feature be processed? Kindly furnish some reference machining solutions.
2	When performing the rough machining of mating surface process on a cylinder head part on a machining centre designated TH6563 × 63A, what tool should be used to complete the operation of drilling a threaded bottom hole, and the corresponding precautions.
3	What is the typical machining accuracy used to rough out the front and rear ends of a cylinder head part on a five-face machining centre? If a face mill is used to machine the ends, what is the recommended speed?
4	others ...

Table 6. Machining process design requirement entities.

Label: Name	Attribute: Value
Parts: Cylinder Head	null: null
Operations: Machining of Reference Surface	null: null
Device: Machining Centre	Device Code: TH6563 × 63A
Features: Blind Hole	null: null

machining process solutions. In contrast, rule knowledge is more scattered and flexible, allowing machining process solution generation better scalability.

4.2. Solution generation

4.2.1. Machining process design requirements

The requirements for machining process design come in various forms, as shown in Table 5. To fulfil the demands of practical machining process design for diesel engine components, let us consider the process requirements put forth by manufacturing personnel: 'During the machining of the reference surface for the cylinder head component on the machining center with the code TH6563 × 63A, how should the blind hole feature be processed? Kindly furnish some reference machining solutions.' Following the requirement identification process outlined in Section 3.2.1, the entities of machining process design requirements are detailed in Table 6 provided below.

The paragraph explores the instantiation of the state space through semantic similarity calculations between demand entities and instances of parts, operations, devices, and features derived from the machining process knowledge graph. Instances with higher similarity scores are strategically chosen to represent the state space instantiation, leading to diverse representations based on distinct label combinations. An illustrative example is presented for the instantiation of process information concerning the cylinder head part, reference surface machining operation, machining centre device, and the blind hole feature, as depicted in Figure 8. The machining process knowledge graph supplements ambiguous process information identified in the proposed process requirements. For example, it supplements intricate details related to feature types, dimensional accuracy, and surface roughness, which were not explicitly stated for the original blind hole feature. This procedure not only aids process designers in elucidating design requirements but also contributes to establishing a more comprehensive initial state space for the deep reinforcement learning model to formulate solutions. The varied representations stemming from

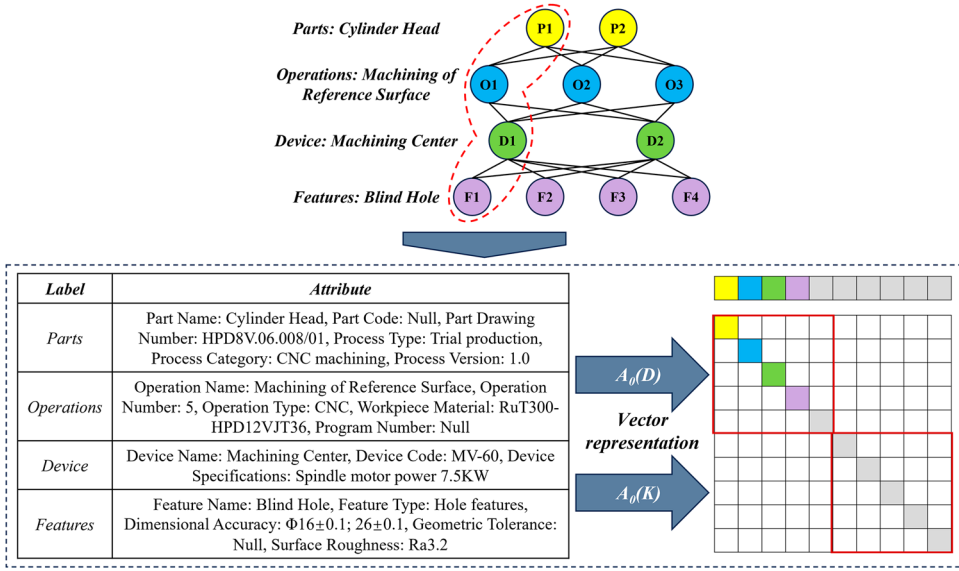


Figure 8. Schematic diagram of the machining process requirement instantiation.

diverse instantiation demands for the initial state space also empower the generated results to encompass a broad spectrum of conceivable scenarios.

4.2.2. Setting up and training of DRL model

Following the framework outlined in Section 3.2.2, we have developed a deep reinforcement learning model. Considering the constraints imposed on the dimensions of the model's state and action spaces by the quantity of elements in the machining process design, as stipulated by the top-level model, and the influence of the number of case and rule knowledge units in the constructed machining process knowledge graph on the model's operational efficiency and iteration steps, the essential hyperparameter settings for the model, necessary for generating low-parameter and small-scale machining process solutions, are delineated below: the learning rate for the *actor* is set at 0.001, the learning rate for the *critic* is 0.002, both the actor and critic neural networks are configured with three hidden layers, each comprising 64 neurons, the *discount factor* is fixed at 0.9, the *batch size* is set to 64, the *maximum step* is 200, $W = [W_1, W_2, W_3] = [1.0, 0.8, 0.6]$, $P = [P_1, P_2, P_3] = [0.8, 0.2, -1.0]$, *factor* = [1.5, 0.5], the *S_dim* is 100, the *A_dim* is 4, and all other parameters are maintained at their default values.

This study's configuration of model state, action, and reward is tailored to the distinctive framework integrating machining process knowledge representation, with DRL serving as the cornerstone for generating machining process design solutions. To assess the efficacy of the chosen the DDPG-based framework, the present section conducts an analysis of the method's suitability across various instantiated demand inputs and juxtaposes its performance against alternative DRL frameworks operating under identical input conditions.

The adaptability of the model to various instantiated requirements reflects its capacity for transferring knowledge to analogous demands and ensuring the stability of the

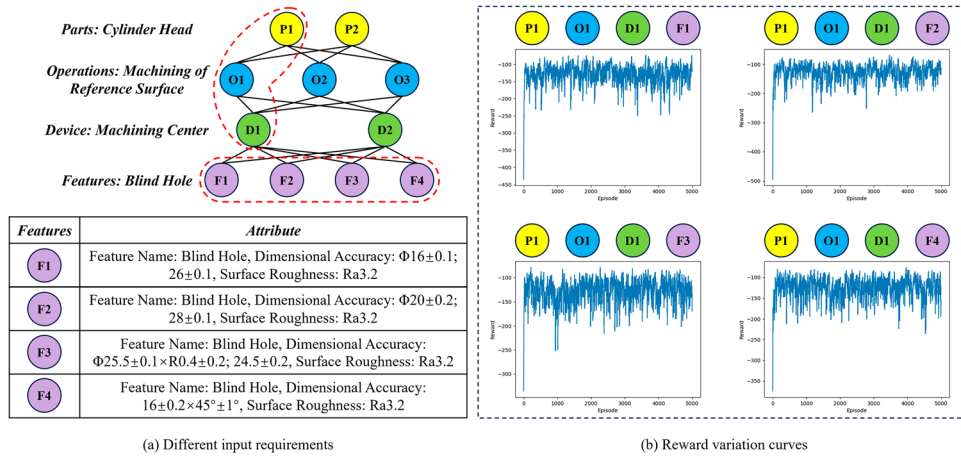


Figure 9. Reward variation curve under different input requirements (partial results).

Table 7. Training time and learning results of different DRL frameworks.

DRL framework	Training time	Best reward
DDPG	628.152s	72.393
A2C	593.264s	50.824
SAC	540.103s	42.855
PPO	530.222s	12.765

solution generation. Each initial state space matrix associated with the proposed process design specifications is independently fed into the DRL model. Illustrated in Figure 9, the total rewards curve corresponding to the four distinct instantiated blind hole features in (a) serves as an exemplar generated through model training, as depicted in (b). Analysis of the training outcomes from the initial 5000 episodes reveals the rapid convergence of model rewards stemming from diverse instantiated requirements. Although occasional fluctuations occur in individual episodes, the overall convergence trajectory of each reward remains broadly consistent. This consistency implies that the approach effectively accommodates a variety of instantiated requirements in machining process design.

To further validate the performance of the model, the DDPG-based framework in this paper is compared with other typical DRL frameworks such as Actor-Critic (A2C), Soft Actor-Critic (SAC), and Proximal Policy Optimisation (PPO). Each framework was trained 10 times based on the same machining process knowledge graph. The cumulative rewards of every 200 steps were treated as episode rewards, and the reward changes were recorded for 500 episodes. The results of the 10 trainings were summarised, and the mean and variance were calculated. F-tests were conducted to check the reward results of different DRL frameworks. The experimental results are shown in Figure 10 and Table 7.

In Figure 10(a), the reward change curves of the DDPG-based framework converge faster than those of the A2C, SAC, and PPO-based frameworks. Additionally, the method in this paper can obtain greater rewards with a slight difference in training time, with the best reward of the DDPG-based framework (72.393), outperforming the A2C-based (50.824), SAC-based (42.855), and PPO-based (12.765) frameworks.

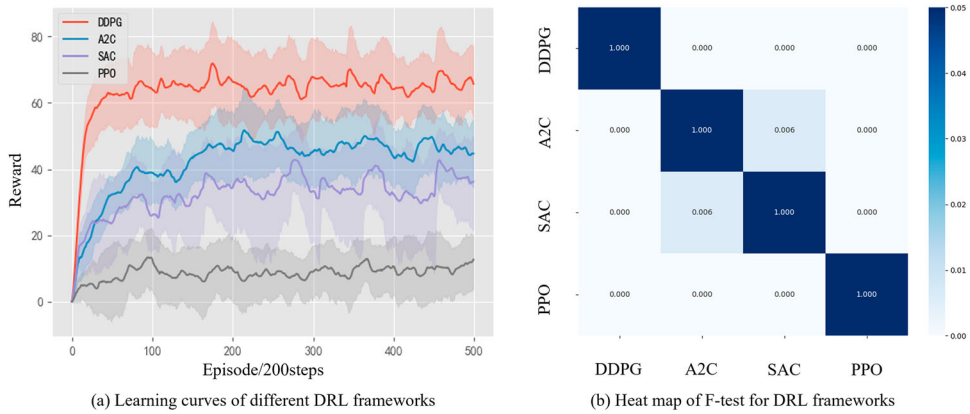


Figure 10. Comparative analysis of different DRL frameworks.

Furthermore, the results of the q-value heat map shown in Figure 10(b) at a significant level of 0.05 also indicate that the performance of the DDPG-based framework is substantial compared to all the other frameworks, as the q-values are all 0.000, which is much less than 0.05. Therefore, the DDPG-based framework demonstrates high performance in generating the design solution for the machining process in this paper.

Compared to the traditional workflow for machining process design, which relies on manual expertise, the method proposed in this paper can significantly diminish the expenses incurred by design personnel through the repetitive adjustment of process solutions. By harnessing the capabilities of DRL to distil and internalise knowledge pertaining to machining processes from a wide array of requirements, this approach delves into the implicit correlations inherent within machining process knowledge, effectively integrating them into the design process. In contrast to optimisation methods tailored exclusively to specific problems, this method exhibits enhanced adaptability to intricate design requirements. Through the synergistic fusion of KG's precision with DRL's reasoning prowess, machining process designers are empowered to procure trustworthy design solutions swiftly.

4.2.3. Analysis of generated solution results

Illustrated by fulfilling the machining process requirement for blind hole processing in the given cylinder head component, the machining process design solution is deduced through the model's reasoning, as depicted in the upper portion of Figure 11. The interpretability advantages inherent in reinforcement learning enable more seamless integration with the pragmatic facets of machining process solution generation tasks. This approach, relying on the framework delineated for solution space, partition, and reward mechanisms, facilitates categorising the solution generation process into two principal segments: the generation of pivotal information and the generation of ancillary resources. In the former segment, the model prioritises generating elements within the D space of machining process solutions, particularly in regions characterised by higher reward weights. Notably, this includes process elements in the D space that, although not explicitly mentioned in the requirements, serve as crucial nodes linking operations and features in the top-level model.

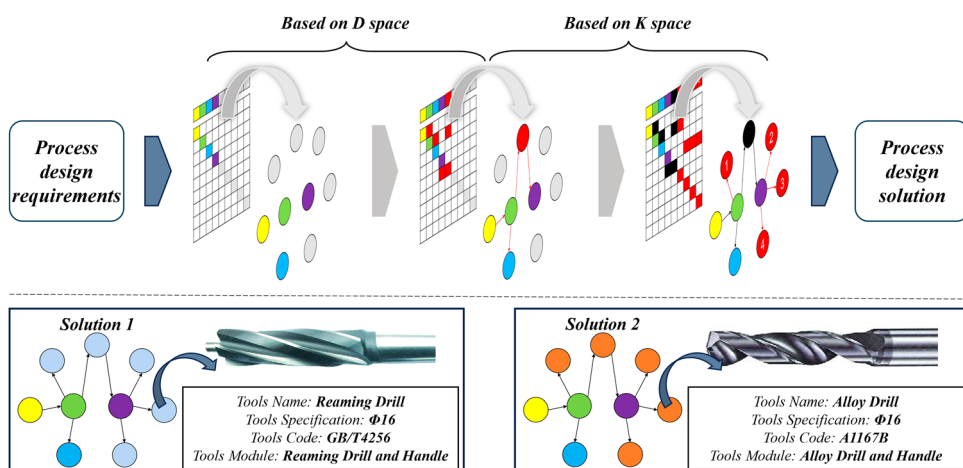


Figure 11. Example of solution generation.

Consequently, process-related cases or rule units gleaned from the machining process knowledge graph receive augmented reward weights. The model precedes sourcing process knowledge with the utmost unit reward when the evaluative judgment is applied to the overall solution generated in each iteration. This strategic emphasis aims to mitigate the potential decrease in overall rewards or even the escalation of penalties arising from belated-generation processes. In the latter segment, the model tends to postpone generating elements in the K space, such as cutting tools and accessories, which are not explicitly outlined in the requirements. This strategic deferral is geared towards eliciting an augmented overall reward, culminating in a final solution that more effectively aligns with the requisites of machining process design.

Notably, this method uniformly represents machining process case knowledge and rule knowledge as knowledge units in triplets. By assigning weights to unit rewards, the rewards for case knowledge units surpass those for rule knowledge units. This characteristic empowers the deep reinforcement learning model in this study with the capacity not only to comprehend the procedural aspects of solution design, initiating from design requisites and prioritising core content, and ultimately complementing additional information, as outlined in the CDK model, but also to discern, when the reward values computed by the average similarity are proximate, the preference for utilising structurally complete yet rigid machining process case knowledge. Subsequently, it contemplates structurally dispersed but adaptable machining process rule knowledge. To a certain extent, this methodology alleviates the inflexibility associated with reusing case-type machining process knowledge and the fragmented reasoning of rule-type machining process knowledge. It capitalises on the advantageous features of case-type knowledge to ensure the overall coherence of solution content while employing the flexibility of rule-type knowledge for localised reasoning. It leads to generating solutions that exhibit enhanced reliability and flexibility in meeting machining process design requirements.

The machining process design solutions exhibit varying contents contingent upon instantiation details, as depicted in the lower section of Figure 11. Process engineers can expeditiously extract insights from the ultimate process plan concerning the strategy for machining blind hole features on a cylinder head component utilising a machining centre

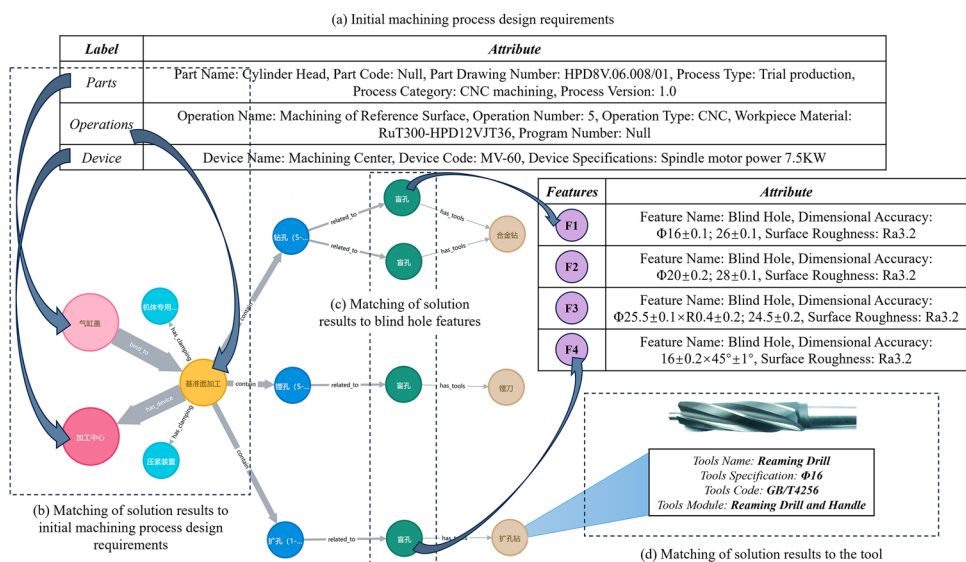


Figure 12. Schematic details of the machining process design solution.

identified by the code TH6563×63A. They are presented with procedural alternatives, including tool GB/T4256, featuring a $\Phi 16$ reaming drill, or tool A1167B, characterised by a $\Phi 16$ alloy drill. The chosen tools align with the size accuracy requisites of the blind hole feature, $\Phi 16 \pm 0.1$, stipulated in the corresponding instantiation demand. In addition to furnishing tooling data aligned with machining process design specifications, several machining process solutions are illustrated in Figure 12. Originating from the example design criteria, these solutions align with the initial requisites and furnish supplementary specialised fixture details tailored to the machining centre. Furthermore, a range of feasible machining steps, encompassing drilling, boring, and reaming, is delineated based on ambiguous blind hole specifications. Through the methodology advanced in this paper, streamlined access to macro-level information encompassing parts, processes, operations, equipment, fixtures, and tools that fulfil machining process design criteria is facilitated. This approach furnishes a resolution for process engineers to promptly acquire efficacious machining process knowledge within structurally diverse and intricate machining knowledge.

5. Conclusions

In conclusion, this paper accomplishes a consolidated representation and inferential reutilisation of historical case knowledge and general rule knowledge within machining processes. The structural diversity of machining process cases and rule knowledge is systematically decomposed into structurally consistent triple knowledge units. This decomposition establishes a foundational representation crucial for the seamless integration of case and rule knowledge during the solution generation. The deep reinforcement learning model is strategically guided by incorporating a reward mechanism, prioritising acquiring the integrity of case knowledge and the adaptability of rule knowledge. This method mitigates, to some extent, the challenges associated with the rigid reuse of case knowledge and the fragmented reasoning of rule knowledge, which are prevalent issues in traditional

approaches. The synergistic utilisation of a knowledge graph and deep reinforcement learning in the intelligent generation of machining process design solutions enables practitioners to efficiently procure process solutions aligning with design requirements. It aids in promptly identifying feasible elements related to machining process design requirements, encompassing critical process information closely associated with requirements and less-critical process resource information. It expedites the initial formulation of design solutions with the reference value. Furthermore, converting the generated solutions into a knowledge graph format, conducive to easy querying and retrieval, lays the groundwork for the rapid accumulation of subsequent machining process solutions.

Indeed, the approach delineated in this paper is not without limitations. The DDPG-based deep reinforcement learning model framework exhibits sensitivity to parameter scale and task environment. Given the abundance of data and complexity tasks inherent in machining process design and other design scenarios, this study has confined its investigation to scenarios with fewer parameters and more minor scales. In the future, our research endeavours will focus on developing models tailored to the intricate demands of the field. This includes exploring the adoption of the twin delayed deep deterministic policy gradient framework, known for its robust training structure, to address the limitations posed by the DDPG concerning parameter scale and task environment. By doing so, we aim to bolster the task-awareness and inference-planning capabilities inherent in the deep reinforcement learning model. Additionally, we will investigate further the representation of knowledge graphs within multimodal process design scenarios. Specifically, we will explore methods for joint representation, integrating requirement text, design drawings, and 3D models, building upon the knowledge representation approach outlined in this paper. Such efforts are intended to enhance the model's proficiency in characterising process knowledge across diverse modalities, thereby facilitating the seamless integration of the universal large language model with process design practices. Ultimately, our goal is to meet the growing demand for intelligent generation of intricate process design solutions and contribute to advancing intelligent process design amidst the evolving landscape of cognitive intelligence trends.

Acknowledgments

This paper is an outcome of the International Systems Realisation Partnership between the Institute for Industrial Engineering @ The Beijing Institute of Technology, China, The Systems Realisation Laboratory @ The University of Oklahoma, USA, the Design Engineering Laboratory @ Purdue, USA, and The Systems Realisation Laboratory @ The University of Liverpool, UK. Thanks for the support of the Key Laboratory of Industry Knowledge & Data Fusion Technology and Application, Ministry of Industry and Information Technology, Beijing Institute of Technology.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Beijing Institute of Technology Research Fund Program for Young Scholars; National Natural Science Foundation of China: [grant no 52105241]; National Ministries Projects of China: [grant no JCKY2020210B007, 50923010601].

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